

INFT 1100 Group 5 Project Deliverable 2

Collaborators:

*Daniel Lindo – Tim Loudon – Melvin Viado – Emmanuel Ajayi – Diksha
Talwar*

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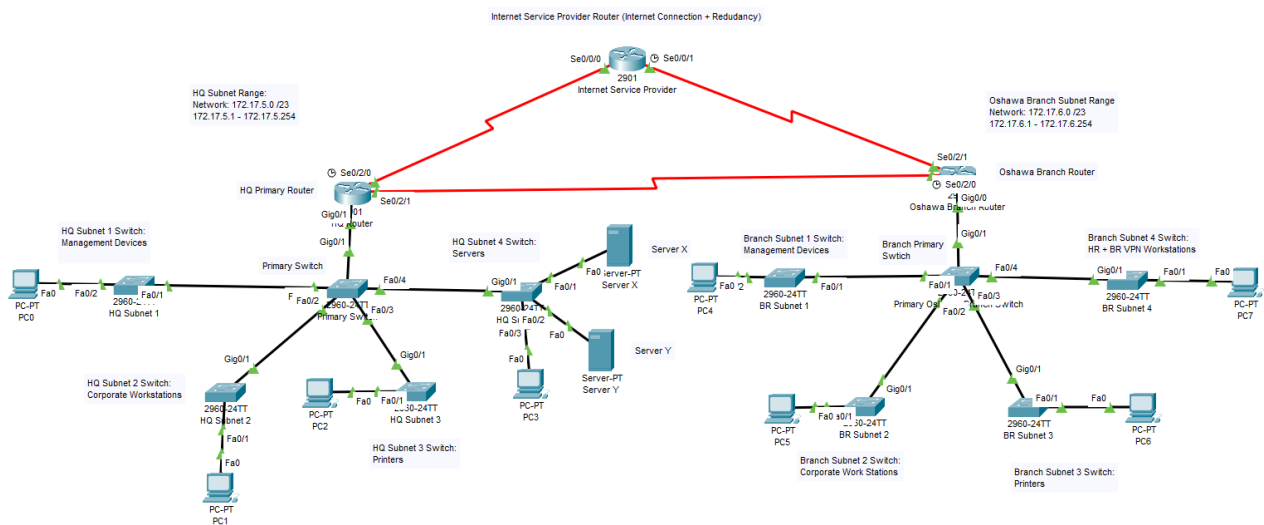
Durham College

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Network Design

Topology:



Topology Modifications:

With a company the size of SecureArc, we wanted to make sure that the network was reliable for employees that needed to communicate information back and forth between the Headquarters and the Oshawa teams. To do this, we implemented 3 routers, one for each of the teams and another for allowing employees access to the internet. By creating the topology this way, there is added redundancy in case one of the connections between routers breaks as they

can still route traffic through the internet router. On the switch side, we also plan on adding EtherChannels for the primary switches to boost network-wide efficiency and redundancy.

By assigning the network devices to their own switches depending on device type (workstations, printers, servers, management devices) we have also made this network easier to scale up as needed. By assigning these device types their own switches, it's easier for the company to keep track of device IP addresses and add new ones to the network more easily. By segregating devices by their type, we can more efficiently provide a better quality of service for the employees of the company due to optimising the connections for each device depending on the switch it's connected to. VLANs will be used on the switches to separate these critical end devices, adding to the network's security and boosting performance. OSPF will also be implemented in the switches to stop loops in the network between each other.

Security is very important when making a network like this. To make this network more secure, we plan on adding firewalls between each of the routers to make sure that the information being passed between them is secure and isn't malicious or will damage systems. To mitigate attacks on our switches, we will implement switch hardening as well as administratively shut down unused ports to make sure they can't be used by attackers.

Strong passwords will be implemented for routers requiring administrator access and if remote access is required, we will be using SSH keys as opposed to unsecure methods such as Telnet. We are also considering the possibility of implementing VPNs to further bolster security.

Location HQ-TonontoBR – Oshawa	End Devices	Network Address/Subnet Mask	First Host Ip Address	Last Host Ip Address	Broadcast Address	Host Space
HQ-Subnet 1- Management Devices	12	172.17.5.0 255.255.255.240/28	172.17.5.1	172.17.5.14	172.17.5.15	Number of Usable Hosts: 14 End Devices: 12
HQ-Subnet 2- Corporate Workstations	124	172.17.5.16 255.255.255.128/25	172.17.5.17	172.17.5.142	172.17.5.143	Number of Usable Hosts: 126 End Devices: 124
HQ-Subnet 3- Printers	14	172.17.5.144 255.255.255.240/28	172.17.5.145	172.17.5.158	172.17.5.159	Number of Usable Hosts: 14 End Devices: 14
HQ-Subnet 4- Servers	28	172.17.5.160 255.255.255.224/27	172.17.5.161	172.17.5.190	172.17.5.191	Number of Usable Hosts: 32 End Devices: 28
BR-Subnet 1- Management Devices	25	172.17.5.192 255.255.255.224/27	172.17.5.193	172.17.5.222	172.17.5.223	Number of Usable Hosts: 32 End Devices: 25
BR-Subnet 3- Printers	30	172.17.5.224 255.255.255.224/27	172.17.6.225	172.17.6.254	172.17.6.255	Number of Usable Hosts: 32 End Devices: 30
BR-Subnet 2- Corporate Workstations	120	172.17.6.0 255.255.255.128/25	172.17.6.1	172.17.6.126	172.17.6.127	Number of Usable Hosts: 128 End Devices: 120
BR-Subnet 4- HR+BR VPN Workstations	125	172.17.6.128 255.255.255.128/25	172.17.7.1	172.17.7.254	172.17.7.255	Number of Usable Hosts: 126 End Devices: 125

IP Addressing Scheme:

Brief Rationale

The initial address space allocated to our group, group number 5, is 172.17.5.0 with a /23 CIDR. The subnetting for SecureArc consists of a variable length subnetting masking with addresses in the range of 172.17.5.0 to 172.17.5.255, then 172.17.6.0 to 172.6.255. The subnets were reordered to better fit the subnets, such that BR-Subnet 3 fits in the 172.17.5.0 to 172.17.5.255 range along with HQ-Subnets 1 to 4 and BR-Subnet 1, but BR-Subnet 2 does not fit. This is due to Br-Subnet 2 having 120 end devices but BR-Subnet 3 only has 30, and with HQ-Subnets 1 to 4 and BR-Subnet 1, there would only be 32 address spaces left.

Security Network Recommendations

Network Enhancements:

Enhancement 1: Implement a Unified Firewall and Intrusion Prevention Framework

i) Description of the Concern

While the network diagram shows routers connecting HQ, the Oshawa branch, and the Internet Service Provider (ISP), there is no explicit indication of a dedicated perimeter or internal firewall beyond “firewalls between routers.” This leaves the organization vulnerable to external and internal lateral movement attacks. Without centralized policy enforcement and deep packet inspection, malicious traffic—such as phishing payloads, worms, or unauthorized data exfiltration—can traverse between subnets or across the WAN link undetected.

ii) Example of Specific Threat or Risk

A man-in-the-middle (MITM) attack or ransomware infiltration could exploit open management ports (e.g., SSH, Telnet, SNMP) on routers or switches. Once a single endpoint is compromised (say, a corporate workstation in HQ Subnet 2), the attacker could pivot laterally through VLANs toward sensitive servers in HQ Subnet 4 or branch HR VPN devices. This exposure is exacerbated without a unified inspection point capable of enforcing segmentation and anomaly detection.

iii) Standards and Framework Support

According to the NIST Special Publication 800-41 Revision 1 (“Guidelines on Firewalls and Firewall Policy”), organizations must “implement layered perimeter defenses, including both network and

host-based firewalls, to enforce boundary protection.” Similarly, ISO/IEC 27033-3:2010 recommends deploying firewalls and intrusion prevention systems (IPS) at network boundaries and critical junctions to control data flows and detect intrusions.

Proposed

Solution:

Deploy a next-generation firewall (NGFW) appliance at both HQ and the branch edge. These devices should:

- Integrate with an Intrusion Prevention System (IPS) for real-time packet analysis.
- Enforce zone-based policies (e.g., HQ internal, server farm, branch WAN, Internet).
- Use application-layer inspection to identify risky traffic such as peer-to-peer or command-and-control beacons.
- Support site-to-site IPsec VPN tunnels between HQ and the Oshawa branch for encrypted traffic over the ISP link.

This enhancement provides not only segmentation but visibility into encrypted flows, aligning with NIST’s Zero Trust architecture model (SP 800-207).

Enhancement 2: Strengthen Access Control and Network Device Hardening

i) Description of the Concern

Although the design mentions strong passwords and disabling unused ports, there is still a broader concern: insufficient centralized authentication and authorization for network device access. Each switch and router appears to rely on local credentials, and no mention is made of multi-factor authentication (MFA) or centralized logging. This limits the organization’s ability to enforce role-based access and track administrative actions across devices.

ii) Example of Specific Threat or Risk

If a network administrator’s credentials were compromised—through phishing or a weak password—an attacker could gain privileged access to core switches or routers, modify configurations, or introduce backdoors. Without centralized authentication or configuration integrity checks, such malicious changes could go unnoticed, compromising both availability and confidentiality.

iii) Standards and Framework Support

The NIST Cybersecurity Framework (CSF), under the *Protect (PR)* and *Detect (DE)* categories, specifically emphasizes Access Control (PR.AC) and Audit Logging (DE.AE) as critical safeguards. Likewise, ISO/IEC 27002:2022, Controls 5.17–5.19, require that system and network

administration be protected by strong authentication, least-privilege principles, and logging of all administrative actions.

Proposed Solution:

- Implement AAA (Authentication, Authorization, and Accounting) via a centralized server (e.g., RADIUS or TACACS+) integrated with Active Directory.
- Enforce multi-factor authentication for SSH logins and router management.
- Apply role-based access control (RBAC) so junior admins cannot modify routing tables or VLAN structures.
- Configure Syslog or SIEM integration for audit trails and configuration monitoring.
- Periodically run Cisco AutoSecure or equivalent hardening scripts to disable legacy protocols (e.g., CDP, HTTP, SNMPv1) and restrict management access to trusted hosts.

These measures create a defensible administrative perimeter, ensuring traceability and adherence to ISO/IEC 27001:2022 A.9 – Access Control and IEEE 802.1X (port-based network access control).

Secure Switch and Router Configuration (Cisco CLI):

Since the network will continue increasing, security misconfigurations may leave internal machines, servers, and data vulnerable to unauthorized access. The subsequent Cisco CLI settings are used to establish the defense-in-depth principles on both switch and router, including authentication, user authorization, the endpoint of connectivity, and traffic paths.

i) Password and Lockout Configuration

Goal: Eliminate unauthorized administrator access to networking devices and use of robust passwords.

```
hostname SecureArc-SW1
enable secret $uperS3cur3Pwd
service password-encryption
line console 0
  login local
  exec-timeout 5 0
  password COnsole@123
line vty 0 4
  login local
  transport input ssh
  exec-timeout 10 0
```



```
username Admin privilege 15 secret N3tw0rk@2025
login block-for 60 attempts 3 within 30
ip domain-name securearc.local
crypto key generate rsa modulus 2048
```

Explanation: These policies are used to secure administrative access through the implementation of the encrypted authentication (SSH, enable secret), password complexity, and account lockout policies. This is in line with the NIST SP 800-53 (AC-2, IA-5) standards.

ii) Port Security Configurations

Port security identifies the MAC address of a device to a particular switch port to minimize the occurrence of rogue devices connection.

(a) Workstation Port

```
interface FastEthernet0/2
description Workstation-PC
switchport mode access
switchport access vlan 10
switchport port-security
switchport port-security maximum 1
switchport port-security violation restrict
switchport port-security mac-address sticky
spanning-tree portfast
```

Reasons: EnforcedOne authorized workstation on a port, Workstation overwhelm: Due to sticky MAC, whitelisting is performed automatically.

(b) Server Port

```
interface FastEthernet0/10
description Server-X
switchport mode access
switchport access vlan 20
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address 00AA.BBCC.DDEE
switchport port-security violation shutdown
spanning-tree portfast
```

Why: This is because Static MAC only gets one legitimate server into this connection any other into the connection creates an automatic shutdown if they attempt to spoof.

(c) Unused Ports

```

interface range FastEthernet0/15 - 24
description UNUSED-PORTS
shutdown
switchport mode access
switchport access vlan 999
switchport nonegotiate

```

Reason: Ports which are not used should be disabled and allocated to VLAN 999 to reduce the insider intrusion or the physical intrusion.

iii) Router Hardening

```

hostname SecureArc-R1
no ip http server
no ip http secure-server
service password-encryption
banner motd ^C

```

```

* Authorized access only!          *
* Unauthorized attempts will be monitored. *

```

```

^C
logging buffered 10000 warnings
no cdp run
no ip source-route
ip cef
security passwords min-length 10
line vty 0 4
login local
transport input ssh
exec-timeout 10

```

Rationale: CIS Cisco IOS Benchmarks recommend the use of password encrypting and the disabling of unnecessary services along with displaying legal banners that make the router attack surface smaller.

iv) Primary and Backup Default Routes

```

ip route 0.0.0.0 0.0.0.0 192.168.1.1 10

```

```
ip route 0.0.0.0 0.0.0.0 192.168.2.1 20
```

Explanation The first path is the primary ISP route, and the second one is a backup route, so that the path will be maintained in case of a failure of the primary route. This is in favor of the CIA Triad Availability pillar.

Summary and Security Rationale

Area	Security Benefit	Industry Reference
Password & Lockout	Prevents brute-force attacks and unwanted logins	NIST SP 800-53 AC-2, IA-5
Port Security	Barricades rogue devices and MAC spoofing	ISO/IEC 27001 A.9
Router Hardening	reduces the attack surface and exposed services.	CIS Benchmarks
Default Routes	uses redundancy to ensure business continuity.	NIST SP 800-115

The structure depicts deep network security by using rigorous authentication, control of device access, hardening of router, and redundancy. These arrangements will result in a secure, reliable, and scalable infrastructure that is in line with the NIST and ISO cybersecurity requirements.

Purchase Recommendations

The following are two separate purchase recommendations that will improve the security and network connectivity at SecureArc.

Recommendation 1: Router and Switch Appliances

Outline of Requirements

Cost effective and scalable routers and switches are important, as these key networking devices are the backbone of SecureArc's network, that should optimally last 4 years before requiring an upgrade. Routers and switches on the premises handle the delivery of packets within the network, and as such are necessary for communications company wide. The selected router and switch to use for the network infrastructure should be reliable, have long lasting warranties, be relatively modern, keep up with modern bandwidths, and support the addition of legacy hardware if needed.

SecureArc's router should be a device that not only has granular configuration but should also support efficient bandwidth speeds greater than 100Mb/s along with routing support for a large number of networking devices and enterprise level routing protocols. Likewise, the switch selected should also support modern bandwidth rates, along with supporting a wide array of enterprise level protocols in tandem with the router, to ensure that both devices offer efficient packet delivery to critical end devices.

Finally, the devices selected should be rack mountable, as this will aid in organizing networking equipment for future additions when needed. PoE capability is also a must, for the addition of security cameras, IP Phones and access points that may be added to the network at a later date. Overall, the selected devices should be dependable, include fair warranties, and most of all have granular configurations especially in relation to scalability. Devices with these requirements would be most optimal for SecureArc's ever-growing network.

Options, Justification, and Procurement

Introduction

When it comes to the devices that could be used to solve our security network solution, networking devices like the Cisco Catalyst 2960-L network switch and Cisco C1111-8P ISP router come to mind. If the Cisco vendor is not preferred by the client, there are also the line of Ubiquiti networking devices, which include devices like the UniFi Switch Lite, and routers like the UniFi Next-Gen Gateway UXG Pro.

Both vendors provide effective network solutions, along with security options, with Cisco having longer experience in the IT industry versus Ubiquiti. In the following sections I will break down the options from both Vendors, and why one may be the preferable option over the other.

Cisco

Switch: Cisco Catalyst 2960-L

When it comes to Cisco, they are an effective brand which showcases consistent results, albeit at a higher price range. Also, Cisco has many lines of devices, I believe the Cisco Catalyst 2960-L network switch, and Cisco C1111-8P ISP router are both solid options for our network solution. Firstly, with the Cisco Catalyst 2960-L switch, each fast ethernet and port can support up to 1Gb in data speeds, with all 24 ports including PoE support (Power over Ethernet).

The switch also supports all the configurations that we have in the included script, with no need to purchase extra software for the switch to run, as the software is included with the purchase, already running and accessible once the switch is powered on. An important distinction is that the PoE model is priced at \$2,200 CAD, while their non-PoE counterpart is cheaper at \$1,200 CAD. The switch supports all enterprise level configurations, such as VLANs, Spanning Tree Protocol, EtherChannel, Access Control Lists, Port Security and much more.

Router: Cisco C1111-8P ISP

For the Cisco C1111-8P ISP Router, this router has many impressive features, such as 8x 1Gb ethernet ports, 2 WAN ports, 1 SFP port, and 4 PoE ports with 2 of them having the capability to power access points, phones and camera. This router supports enterprise protocols, such as OSPF, EIGRP, RIP, static routes, and even supports zone-based firewalls and intrusion prevention. This router is typically priced at \$1400 CAD but can be found at \$850 CAD if purchased used or refurbished. As stated earlier, the current configurations that we have are also supported by Cisco and would make designing and building the network an efficient process. It is also worth noting that both networking devices are rack mountable.

Bonus

Both Cisco devices offer a lifetime warranty; 90 days for software coverage, and lifetime (if you own the item) for hardware coverage. After consulting their 24/7 customer service, your device will be replaced with a new or refurbished but tested version of your damaged hardware.

Ubiquiti

Switch: UniFi Switch Lite

This switch by Ubiquiti has 24x 1Gb ethernet ports, 2x 1Gb SFP uplinks for Fiber Optic, is rack mountable and fan-less which is useful for noise reduction. For protocols, this switch supports VLAN deployment, Spanning Tree protocol, LACP, all at a price of \$300-\$400 CAD depending on the seller. This switch is cheaper and highly efficient, however it is not configurable with cisco commands, instead relying on their proprietary GUI interface, which although useful, may be limited in the complexity of available configurations, however it is still a strong option. The software for this switch is included in the purchase, and the switch also comes with a 2-year warranty if purchased directly from the Ubiquiti store.

Router: UniFi Next-Gen Gateway UXG Pro

When it comes to the UniFi Next-Gen Gateway UXG Pro, this router has also been designed with ease of use being the primary goal. It is also configurable by an easy-to-use GUI, which is also a bit limited in configuration potential due to the GUI reliance and lack of line-by-line commands. This router is powerful, supporting a bandwidth of 10Gbps through its 1 wan port and 1 ethernet port. This router is also rack mountable, with a size of 1RU (1 Rack Unit) and it also supports most of the routing protocols that the Cisco C1111 router supports. Finally, this router has a price range of \$700 - \$1,000 depending on the store selling the item. The software for this router is included with the purchase, and router also comes with a 2-year warranty if purchased directly from the Ubiquiti store.

Recommended Vendor: Cisco

Overall, although Ubiquiti's line of products is cheaper, easier to configure, and more than comparable in device specifications, Cisco takes the lead in the granularity and potential complexity of command configurations with the command line interface, versus Ubiquiti's cloud-based GUI interface, which also requires that you can have a Ubiquiti account. Although Cisco having higher price ranges, takes the advantage in protocol support, configuration granularity, along with device warranties and customer service.

Influence on Workflows

By replacing the routers and switches in the network and upgrading them with newer and more powerful ones, employees of SecureArc will be able to communicate with the other employees and servers at both locations much faster. By implementing these improved devices, the connections will also be more reliable mitigating the risk of outages and will help SecureArc in their efforts to scale up the business.

Newer routers and switches can also be more secure if configured correctly, protecting the company against cyberattacks targeted at these devices. By ensuring that the hardware devices used are up-to-date with modern standards, the entire network becomes faster, efficient, and safer for the employees of SecureArc to use while providing them with the option of easier scalability as they expand their primary location in Toronto and move into a secondary location in Oshawa.

Adverse Consequences

Some adverse consequences of upgrading the routers and switches are the time to complete the upgrade process and potential security vulnerabilities that could arise during the upgrade process.

The time to install the new routers and switches is the most immediate concern facing SecureArc. Due to installation workers needing to take the old devices offline before implementing and configuring the new ones, SecureArc employees and customers could face service outages during this time. This is a necessary flaw however as without implementing the new devices, none of the benefits from them could be gained.

A security vulnerability possible during the installation process is that if the new routers and switches are misconfigured, malicious actors could breach the system and exfiltrate sensitive data by attacking the network through various means such as an overflow attack. It is crucial that SecureArc confirms that their systems are secure after the installation process, possibly through an audit. Attackers could also attack the network during the window between an old router being offline and the new one not set up yet. SecureArc could compensate for this by blocking all external traffic onto the network during this time.

An additional consequence that is not an issue for SecureArc due to their limitless budget but for other companies is cost. It can be quite expensive to purchase the necessary components and costs of labour when scaling up a business and opening a new location that needs to communicate with the main headquarters. It's something to keep in mind but not applicable to this situation.

Recommendation 2: Next-Generation Firewall Appliances

Outline of Requirements

Why is this a requirement?

A next-generation firewall is a requirement for SecureArc because the company is internet-facing, growing fast, and handles valuable digital assets in a weak security environment. A basic firewall will sufficiently protect their public website, artwork database, and internal windows domain as they scale from 500 users and a second site.

The main revenue system that SecureArc runs on is their website and the artwork stored on their database. They also hold sensitive data such as art assets, customer information, and transaction records. They need a solution to reduce website compromise, data theft, or ransomware. A successful attack on a SecureArc server will stop sales and damage customer trust.

SecureArc needs extensive security with their plans to increase the number of employees and to open a second office. There will be more endpoints, administrators, attack surface, and need for security. SecureArc especially needs NGFW-grade controls for remote access on Server X.

What advantages does it bring to the company?

The purchase of a next-generation firewall will grant the company an intrusion prevention system, web filtering, advanced threat protection, application control, threat intelligence and site-to-site VPN for its approximate 500 users and one additional branch.

NGFWs also reduces lateral movement risks by using a combination of network segmentation, deep packet inspection, and threat prevention. If one segment is compromised, network segmentation between user personal computers, individual servers, and guest wireless connections prevents lateral movement.

Next-generation firewalls for small to medium businesses also provide secure gateway features such as malware and phishing blocking, domain name service security, software-defined wide area network, and integrated Wi-Fi or 5G options. NGFWs also can extensively analyze network traffic while identifying and preventing modern cyber threats

For what reason is this purchase recommendation top-priority?

The features of a next-generation firewall is a top-priority purchase recommendation because it provides advanced and comprehensive protection tailored for the modern cybersecurity challenges that small business faces. An NGFW is important for SecureArc because they are fast growing, internet-facing, and hold valuable artwork assets. A next-generation firewall is a necessary foundation to securely scale its operations, protect its sensitive assets, mitigate cyber threats, and provide secure remote and branch connectivity.

Options, Justification, and Procurement

FortiGate by Fortinet

FortiGate has a single appliance that provides NGFW features such as intrusion prevent, application control, web filtering, antivirus, and SSL inspection. These features can help SecureArc with protecting their public web server, windows infrastructure, and artwork databases. FortiGate can tailor to the small to medium business scale of SecureArc and can enforce rules, implement segmentation, and reduce the risk of website compromise and lateral movement.

Check Point Quantum Spark

Check Point Quantum Spark is a set of next-generation firewalls designed for small and medium-sized businesses that provide threat protection in compact, easy-to-deploy appliances. These NGFWs provide firewalling, VPN, antivirus, application control, URL filtering, email security, intrusion prevention, and zero-day protection in an “all-in-one” security gateway.

Choice and Comparison

Although both NGFW services provide security features for small and medium-sized businesses, FortiGate is the choice for SecureArc due to its cost-effective and scalable nature. FortiGate has competitive pricing for their advanced features while Quantum Spark is more expensive and more complex to manage than FortiGate. Additionally, FortiGate is easier to use with their user-friendly interfaces and simpler licensing models, which is great for smaller IT teams.

Influence on Workflows

Network Traffic Alterations

Implementing NGFWs would introduce several operational and technical changes throughout SecureArc's environment. These changes would enhance security and scalability, but they would also influence day-to-day workflows for employees, IT staff, and customers.

NGFW deep packet inspection feature may increase latency, especially during peak hours, because packets are analyzed for malware, application signatures, intrusion attempts, and encrypted traffic patterns. Traffic will be categorized by application and non-business traffic may be blocked or throttled, which could reduce unnecessary bandwidth usage but may initially disrupt user habits.

As part of deploying NGFWs, SecureArc would likely segment their flat network into multiple VLANs such as staff devices, servers, point-of-sale, and guest Wi-Fi. This segmentation changes how traffic flows between departments and servers and requires new routing paths managed through the NGFW.

An NGFW includes VPN capabilities that will enable employees to securely work from home. This introduces increased encrypted traffic flowing through the firewall.

User Procedure Adjustment

NGFW-integrated identity services would require staff to authenticate using MFA for VPN access or sensitive systems. This adds a new step to login procedures. With application control and web-filtering enforced by the NGFW, employees may experience blocks on non-work websites or cloud platforms they previously used freely.

Staff will be able to access File Sharing and Domain Controller services from home as NGFW enables secure VPN connections. Employees must learn basic VPN login procedures, and IT must distribute VPN clients or provide browser-based access. Users may need to request access to certain websites, ports, or applications if those are blocked by new security policies. This introduces more formalized IT processes.

Potential Impacts on Customers

The NGFW's IDS/IPS features protect Server Y (the public website and artwork database) from DDoS attempts, SQL injection, and other attacks which increases website uptime and protects customer data.

Because inbound traffic to the public servers also passes through the NGFW, the firewall must be tuned to avoid false positives that could unintentionally block legitimate customer traffic. Initial tuning may temporarily impact customer access until rules mature. Customers who purchase digital artwork will benefit from improved protection of personal and financial data, building trust and brand credibility.

Adverse Consequences

Even with an unlimited budget, NGFW implementation introduces increased operational complexity. NGFWs are significantly more complex to configure and maintain than traditional firewalls. Firstly, NGFWs require skilled administrators to manage policies, signatures, encryption rules, and segmentation. Secondly, NGFWs require continuous monitoring and updating of intrusion-prevention rules. Lastly, NGFWs require time-intensive initial configuration, testing, and ongoing policy refinement.

During rollout, misconfigured rules could block legitimate traffic, interrupt access to servers, or temporarily disrupt business operations. IT staff may require dedicated training or additional hires to manage the new security platform effectively.